Supply Chain Data Analysis Project

System Analysis & Design Documentation

Project at a glance	
Data	One CSV file with 180,519 rows and 53 columns, plus a data dictionary
Period	Orders from 01 Jan 2015 to 31 Jan 2018
Output	A dashboard
Tools	Excel, Power BI, Python, SQL (for queries only)
Users	Sales, Logistics / Delivery Manager, Board Viewer

Everything in this document comes from the dataset, its data dictionary and the decisions we made about the project. The design parts (pages, KPIs, diagrams) are our suggested design. The original data was not changed.

1. About the Data

1.1 The dataset at a glance

Item	Value
Rows (one row = one order item)	180,519
Columns	53
Orders	65,752
Customers	20,652
Products	118
Categories / Departments	51 / 11
Items per order	1 to 5 (average 2.7)
Duplicate rows	0

1.2 The columns

Meanings come from the data dictionary. "Missing" is the number of empty cells.

#	Column	Type	Missing	What it means
1	Type	Text	0	Type of transaction made
2	Days for shipping (real)	Number	0	Actual shipping days of the purchased product
3	Days for shipment (scheduled)	Number	0	Days of scheduled delivery of the purchased product
4	Benefit per order	Number	0	Earnings per order placed
5	Sales per customer	Number	0	Total sales per customer made per customer
6	Delivery Status	Text	0	Delivery status of orders: Advance shipping , Late...
7	Late_delivery_risk	Number	0	Categorical variable that indicates if sending is late...
8	Category Id	Number	0	Product category code
9	Category Name	Text	0	Description of the product category
10	Customer City	Text	0	City where the customer made the purchase
11	Customer Country	Text	0	Country where the customer made the purchase

#	Column	Type	Missing	What it means
12	Customer Email	Text	0	Customer's email
13	Customer Fname	Text	0	Customer name
14	Customer Id	Number	0	Customer ID
15	Customer Lname	Text	8	Customer lastname
16	Customer Password	Text	0	Masked customer key
17	Customer Segment	Text	0	Types of Customers: Consumer , Corporate , Home Office
18	Customer State	Text	0	State to which the store where the purchase is registered...
19	Customer Street	Text	0	Street to which the store where the purchase is...
20	Customer Zipcode	Number	3	Customer Zipcode
21	Department Id	Number	0	Department code of store
22	Department Name	Text	0	Department name of store
23	Latitude	Number	0	Latitude corresponding to location of store
24	Longitude	Number	0	Longitude corresponding to location of store
25	Market	Text	0	Market to where the order is delivered : Africa , Europe...
26	Order City	Text	0	Destination city of the order
27	Order Country	Text	0	Destination country of the order
28	Order Customer Id	Number	0	Customer order code
29	order date (DateOrders)	Date (text)	0	Date on which the order is made
30	Order Id	Number	0	Order code
31	Order Item Cardprod Id	Number	0	Product code generated through the RFID reader
32	Order Item Discount	Number	0	Order item discount value
33	Order Item Discount Rate	Number	0	Order item discount percentage
34	Order Item Id	Number	0	Order item code
35	Order Item Product Price	Number	0	Price of products without discount
36	Order Item Profit Ratio	Number	0	Order Item Profit Ratio
37	Order Item Quantity	Number	0	Number of products per order
38	Sales	Number	0	Value in sales
39	Order Item Total	Number	0	Total amount per order
40	Order Profit Per Order	Number	0	Order Profit Per Order
41	Order Region	Text	0	Region of the world where the order is delivered : ...
42	Order State	Text	0	State of the region where the order is delivered
43	Order Status	Text	0	Order Status : COMPLETE , PENDING , CLOSED ,...
44	Order Zipcode	Number	155,679	Not in the data dictionary
45	Product Card Id	Number	0	Product code
46	Product Category Id	Number	0	Product category code
47	Product Description	Number	180,519	Product Description
48	Product Image	Text	0	Link of visit and purchase of the product
49	Product Name	Text	0	Product Name
50	Product Price	Number	0	Product Price
51	Product Status	Number	0	Status of the product stock :If it is 1 not available , 0...
52	shipping date (DateOrders)	Date (text)	0	Exact date and time of shipment
53	Shipping Mode	Text	0	The following shipping modes are presented : Standard...

1.3 Main categories

Column	Values
Type	DEBIT, TRANSFER, PAYMENT, CASH
Delivery Status	Late delivery, Advance shipping, Shipping on time, Shipping canceled
Shipping Mode	Standard Class, Second Class, First Class, Same Day
Customer Segment	Consumer, Corporate, Home Office
Market	LATAM, Europe, Pacific Asia, USCA, Africa
Order Status	COMPLETE, PENDING_PAYMENT, PROCESSING, PENDING, CLOSED, ON_HOLD, SUSPECTED_FRAUD, CANCELED, PAYMENT_REVIEW
Late_delivery_risk	0 = not late, 1 = late

1.4 Data quality issues

These are the problems we found. We did not fix or delete anything in the original file.

Issue	Details
Empty column	Product Description is empty in every row.
Mostly empty column	Order Zipcode is missing in 86.2% of rows and is not in the data dictionary.
Constant columns	Product Status is always 0. Customer Email and Customer Password have the same masked value in every row.
A few missing values	Customer Lname (8 rows) and Customer Zipcode (3 rows).
Shifted values	3 customers have City = "CA" and a ZIP code in the State column.
Mixed languages	Some country and place names are in Spanish (for example "EE. UU.").
Extra spaces	A few labels end with a space (for example in Order Region and Department Name).
Same name, two IDs	The category "Electronics" has two Category Ids (13 and 37).
Discount does not match the rate	In 10,029 rows the rate is 0.06 but the discount is about 5.5% of Sales.
Negative profit	33,784 rows (18.7%) have a negative Order Profit Per Order.
Dates are text	Order and shipping dates are stored as text (M/D/YYYY H:MM).

1.5 Duplicate and calculated columns

These pairs of columns have the same values in every row:

Column	Same as
Order Customer Id	Customer Id
Product Category Id	Category Id
Order Item Cardprod Id	Product Card Id
Order Item Product Price	Product Price
Benefit per order	Order Profit Per Order
Sales per customer	Order Item Total

Some columns can be calculated from others, and the data follows these rules:

Rule	Result in the data
Sales = Order Item Product Price x Order Item Quantity	Always true
Order Item Total = Sales - Order Item Discount	Always true
Late_delivery_risk = 1 when Delivery Status is "Late delivery"	Always true
Delivery Status follows real days vs scheduled days	True for all orders that were not cancelled

1.6 How the data is organized

Each row is one order item. An order can have 1 to 5 items. All order columns (dates, status, shipping, destination) are the same for every item of the same order, and all customer columns are the same for every order of the same customer. The same is true for products, categories and departments. This is why the data can be split into six tables (Section 3).

Identifier	Distinct values	Note
Order Item Id	180,519	Unique in every row
Order Id	65,752	Repeats for each item in the order
Customer Id	20,652	Repeats for each order
Product Card Id	118	
Category Id	51	Each category belongs to one department
Department Id	11	

2. Problem Statement & Objectives

2.1 Problem statement

This project analyzes a static supply chain dataset of 180,519 order items (orders from 2015 to January 2018). The data covers sales, discounts, profit, shipping and delivery status. We build a dashboard that shows this information to three types of users: Sales, Logistics / Delivery Manager and Board Viewer.

2.2 Objectives

#	Objective
1	Check the data and record every quality issue without changing the original values.
2	Show Sales users the sales, discount and profit.
3	Show the Logistics / Delivery Manager how shipping and delivery are performing.
4	Give the Board Viewer a simple overview of the main KPIs.

2.3 Scope

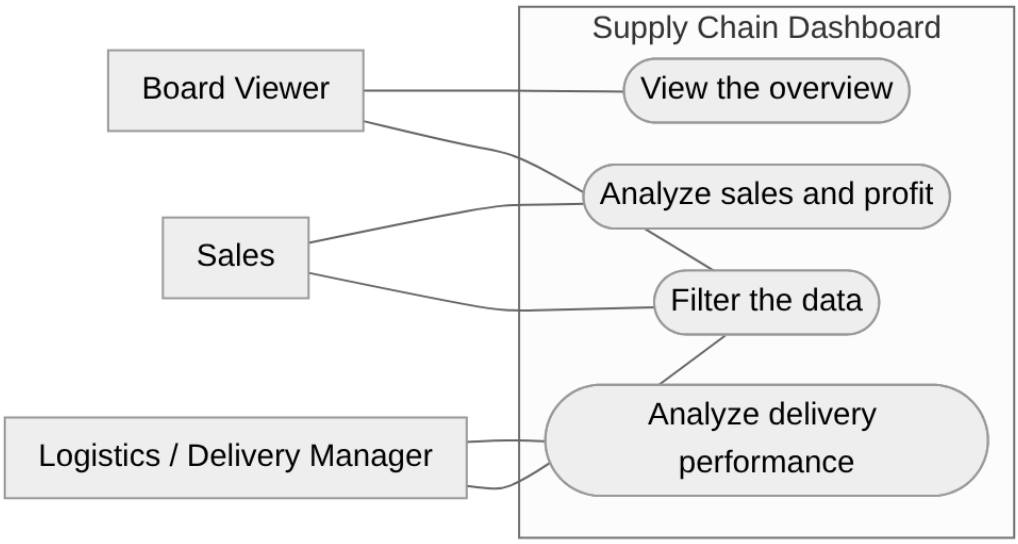
In scope	Out of scope
One static dataset and its data dictionary	Building a database (SQL is used only to run queries)
A dashboard with 3 pages	Updating the data automatically (the file is static)
Tools: Excel, Power BI, Python, SQL	

2.4 Users

User	What they need	Data used
Sales	Sales, discount and profit	Sales, Order Item Discount, Order Profit Per Order, Customer Segment, Category Name
Logistics / Delivery Manager	Shipping time and late deliveries	Days for shipping (real), Days for shipment (scheduled), Delivery Status, Shipping Mode
Board Viewer	A quick overview (view only)	Main KPIs by Market and by time

2.5 Use case diagram

What it shows: Which user does what in the dashboard.



Each user has one main function, and all three can filter the data.

Use case	User	What happens
View the overview	Board Viewer	Open page 1 and read the KPI cards and charts
Analyze sales and profit	Sales	Open page 2 and review sales, discount and profit
Analyze delivery performance	Logistics / Delivery Manager	Open page 3 and review late deliveries and shipping days
Filter the data	All users	Pick filter values and all visuals update

2.6 Requirements

Functional requirements

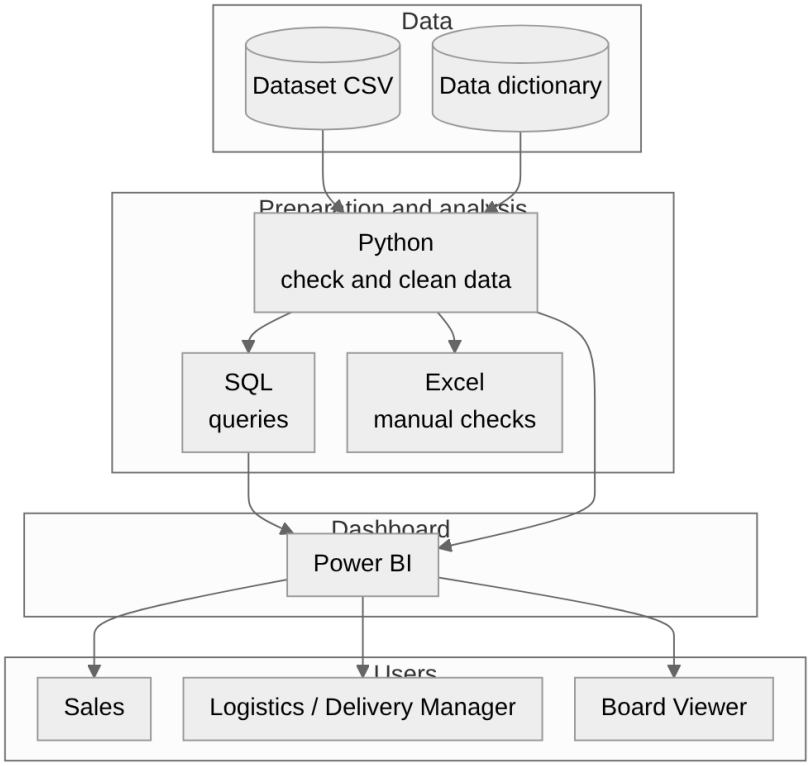
ID	Requirement
FR1	Load the CSV file without changing the original values.
FR2	Check the data and compare it with the data dictionary.
FR3	Calculate the KPIs with SQL queries.
FR4	Page 1: overview for the Board Viewer.
FR5	Page 2: sales and profit for Sales.
FR6	Page 3: delivery performance for the Logistics / Delivery Manager.
FR7	Filters that update all visuals.
FR8	Check that dashboard totals match the query results.

Non-functional requirements

ID	Requirement
NFR1	Accuracy: every KPI is checked against a second calculation.
NFR2	No original value is changed without being recorded.
NFR3	Easy to use: each user finds their page and filters without help.
NFR4	Repeatable: the same steps give the same results on the same file.

2.7 System architecture

What it shows: The parts of the system and how data moves through them.



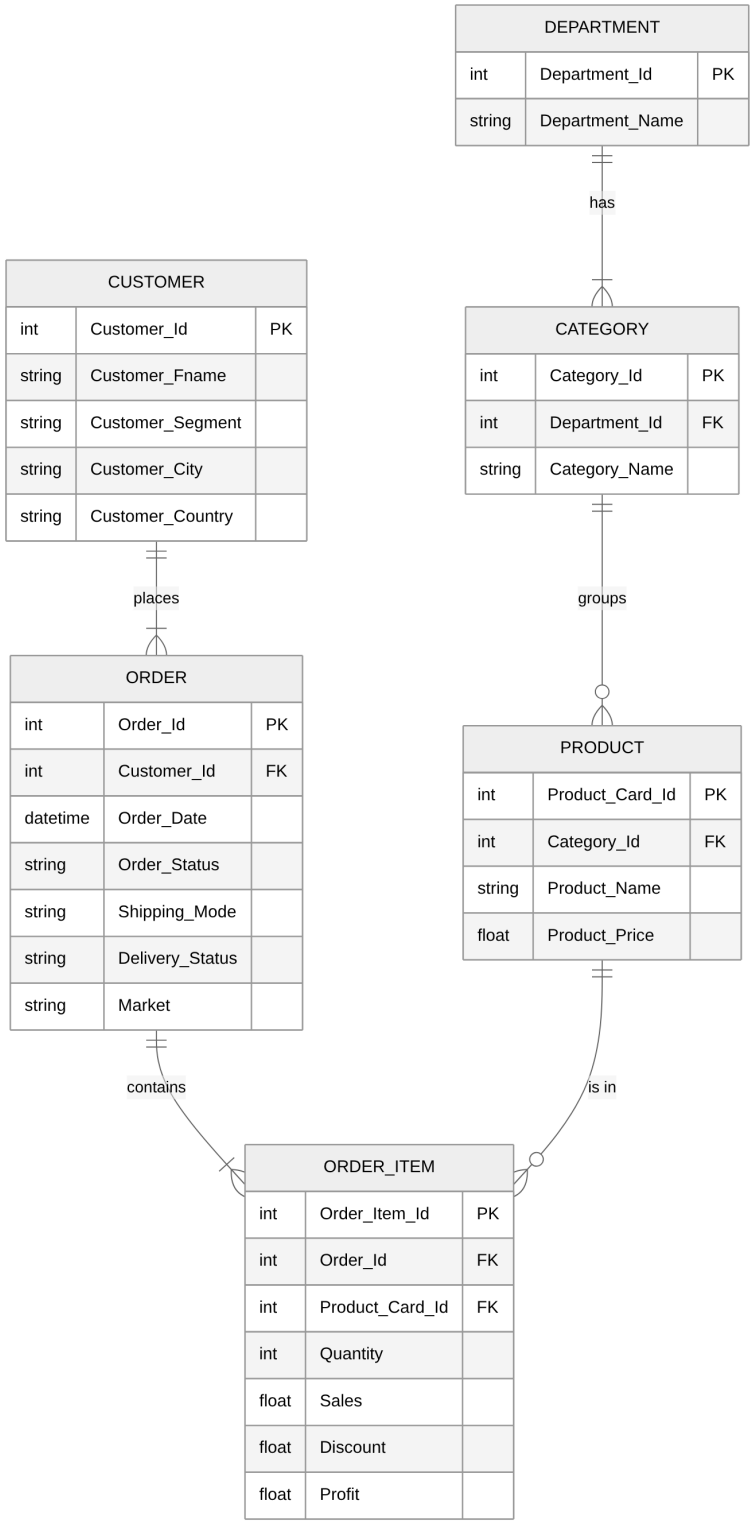
Data goes from the CSV file to Python, then to SQL and Power BI, which show it to the three users. Excel is used for manual checks. There is no database or web application in this project.

3. Database Design

We are not building a database, so only the logical design is needed. It splits the single flat file into six tables. The keys below are chosen because their values are unique or consistent in the data.

3.1 ER diagram

What it shows: the six tables and how they connect.



One customer places many orders, and one order contains one to five order items. A product can appear in many order items. A category groups many products, and a department has many categories.

3.2 Tables

Table	Key	Main columns
CUSTOMER	Customer Id	Customer Fname, Customer Lname, Customer Segment, Customer Street, City, State, Zipcode, Country, Latitude, Longitude
ORDER	Order Id	Customer Id, order date, shipping date, Type, Order Status, Shipping Mode, Days for shipping (real), Days for shipment (scheduled), Delivery Status, Late_delivery_risk, Order City, State, Country, Region, Market, Order Zipcode
ORDER_ITEM	Order Item Id	Order Id, Product Card Id, Quantity, Order Item Product Price, Discount, Discount Rate, Sales, Order Item Total, Profit Ratio, Order Profit Per Order
PRODUCT	Product Card Id	Category Id, Product Name, Product Price, Product Image
CATEGORY	Category Id	Department Id, Category Name
DEPARTMENT	Department Id	Department Name

The duplicate columns from Section 1.5, the empty column (Product Description) and the constant column (Product Status) are left out of the tables.

3.3 Relationships

Relationship	Type	Proof in the data
Customer to Order	One to many	Each order has exactly one customer
Order to Order Item	One to many	Each order has 1 to 5 items
Product to Order Item	One to many	Each item has exactly one product
Category to Product	One to many	Each product has one category
Department to Category	One to many	Each category has one department

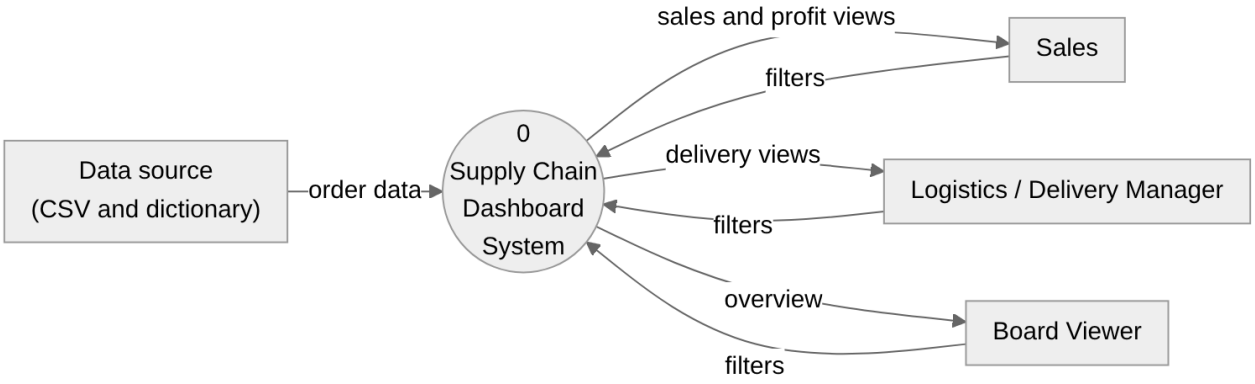
3.4 Normalization

In the original file, order, customer and product details are repeated on every item row. Splitting the data into six tables removes this repetition: each fact is stored once and linked by its key.

4. Data Flow

4.1 DFD Level 0

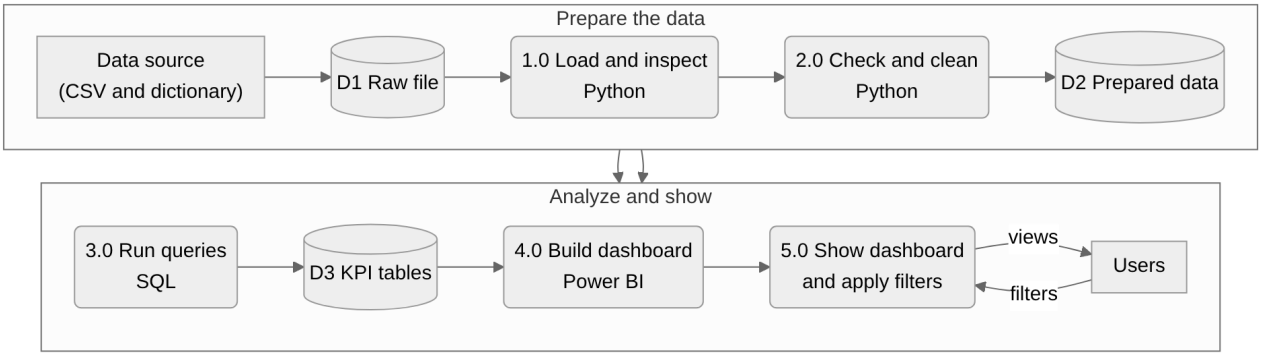
What it shows: The whole system as one process, with the people and data that go in and out.



The data source sends the order data in. Each user sends filters and receives the view they need.

4.2 DFD Level 1

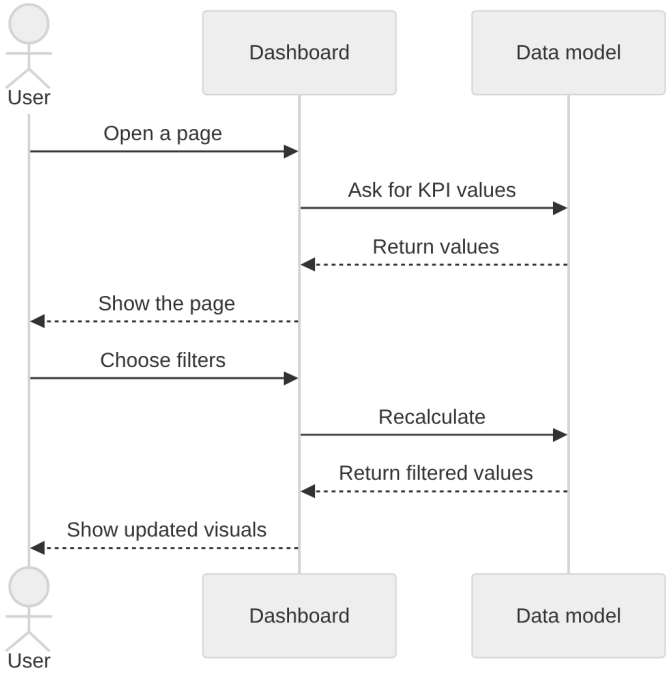
What it shows: The system split into its five main processes and three data stores.



First the raw file is loaded, checked and cleaned (1.0 and 2.0). Then SQL builds the KPI tables (3.0), Power BI builds the dashboard (4.0), and the dashboard is shown to the users (5.0).

4.3 Sequence diagram

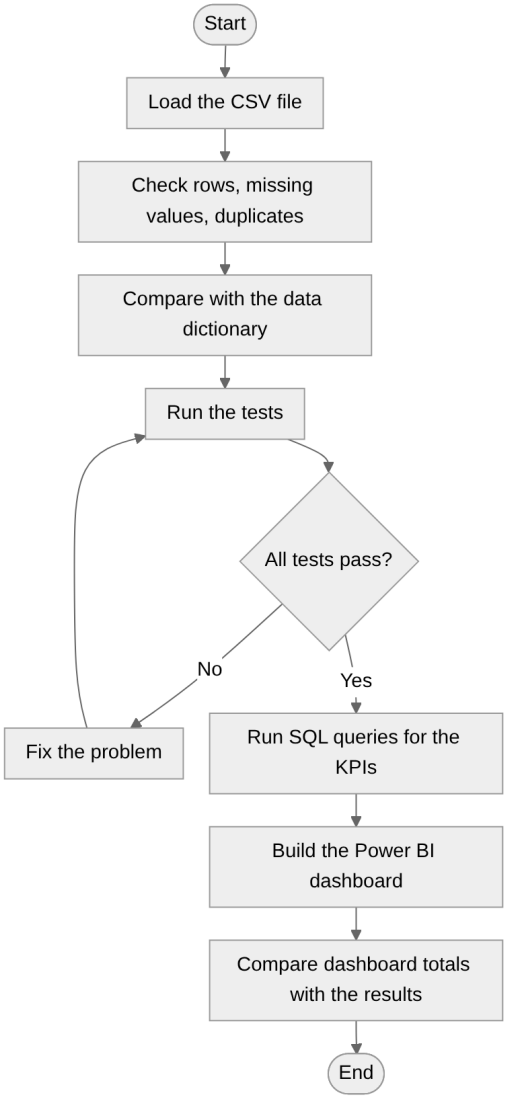
What it shows: What happens when a user opens a page and uses a filter.



The dashboard asks the data model for values and shows them. When the user picks a filter, the values are recalculated and the visuals update.

4.4 Activity diagram

What it shows: The steps to prepare the data and build the dashboard.



The data is loaded and checked, and the tests must pass before the KPIs are calculated and the dashboard is built.

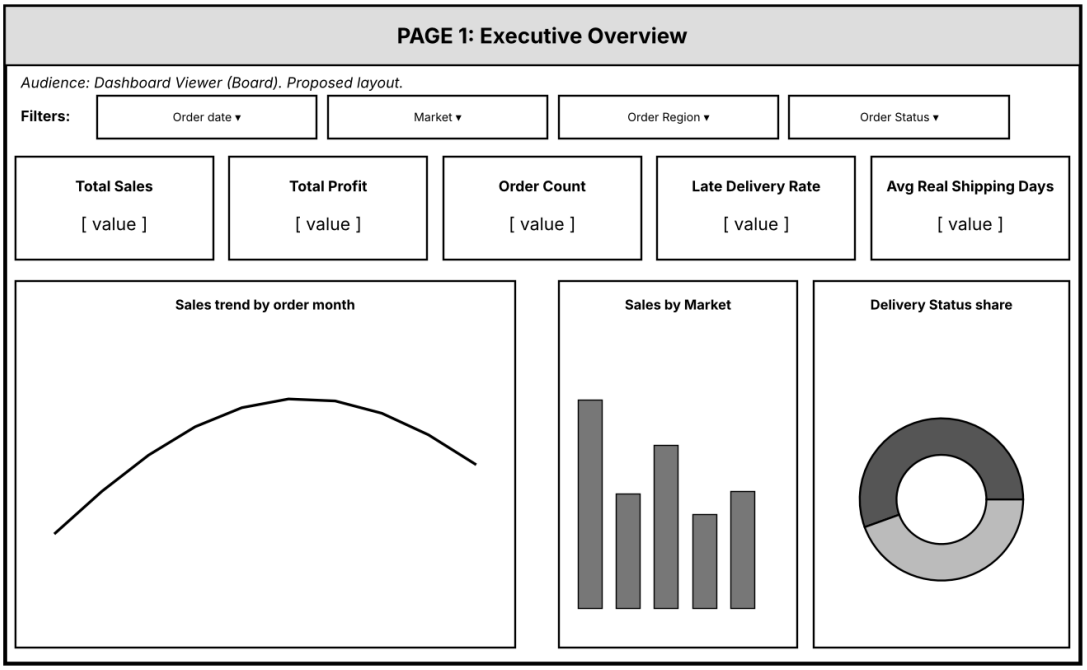
5. Dashboard Design

5.1 Pages

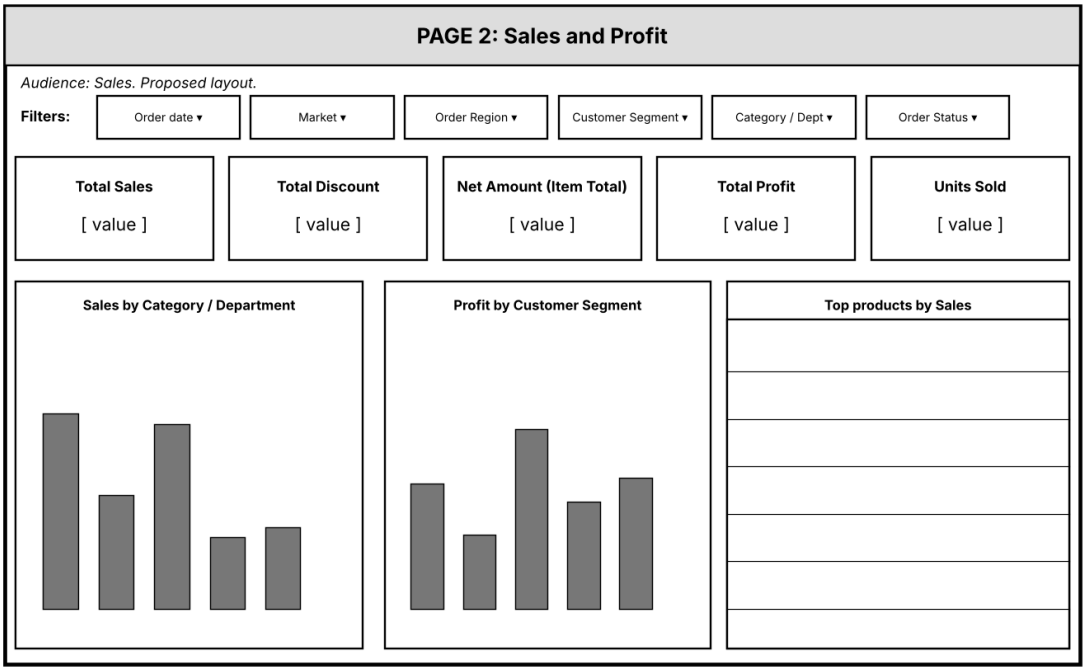
Page	For	Content
1 Executive Overview	Board Viewer	KPI cards, sales trend, sales by market, delivery status
2 Sales and Profit	Sales	KPI cards, sales by category, profit by customer segment, top products
3 Delivery Performance	Logistics / Delivery Manager	KPI cards, delivery status by shipping mode, late rate by market and by month

5.2 Wireframes

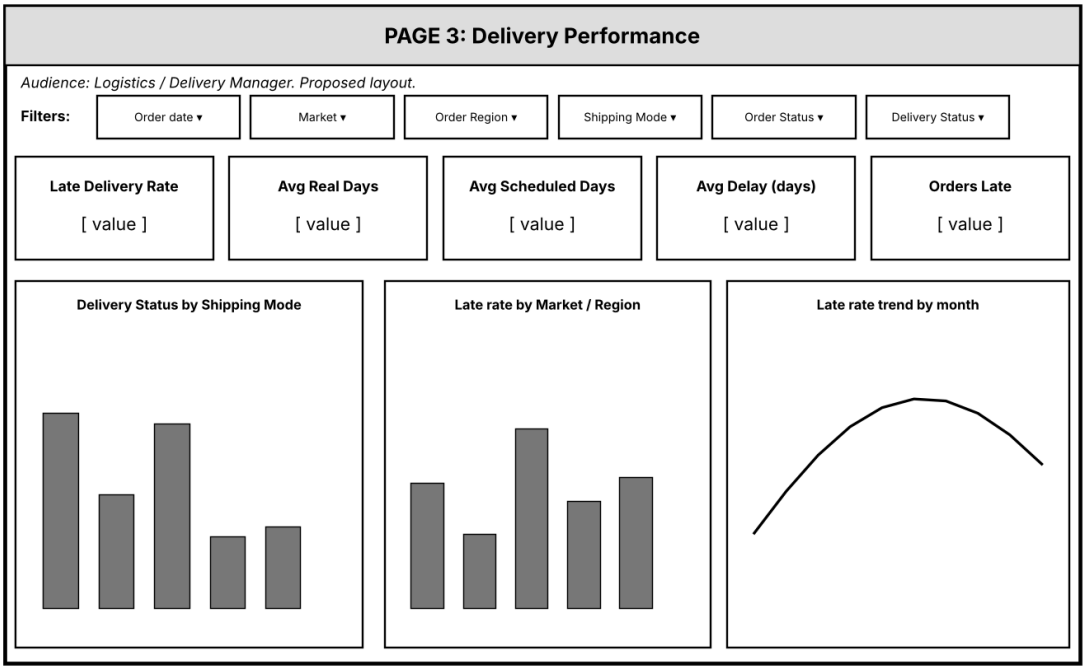
The values are placeholders. The charts show where each visual goes.



Page 1: Executive Overview



Page 2: Sales and Profit



Page 3: Delivery Performance

5.3 KPIs

Each KPI uses columns that exist in the data. The values are calculated on the original file.

KPI	How it is calculated	Value
Total Sales	Sum of Sales	36,784,735.01
Total Discount	Sum of Order Item Discount	3,730,378.40
Net Amount	Sum of Order Item Total	33,054,402.38
Total Profit	Sum of Order Profit Per Order	3,966,902.97
Orders	Count of different Order Id	65,752
Late Delivery Rate	Share of rows with Delivery Status "Late delivery"	54.83%
Average real shipping days	Average of Days for shipping (real)	3.50
Average scheduled days	Average of Days for shipment (scheduled)	2.93
Average delay (days)	Real days minus scheduled days	0.57

The late delivery rate counts every row, including those with Delivery Status "Shipping canceled".

5.4 Filters

Order date, Market, Order Region, Order Country, Customer Segment, Category / Department, Shipping Mode, Order Status and Delivery Status. When a user picks a value, all KPIs and charts on the page update.

5.5 Design tips

Tip	Why
One page for each user	Everyone sees what they need first.
KPI cards on top, filters in one bar, details below	The layout is the same on every page.
Use the same names as the data dictionary	"Sales" and "Order Item Total" are different, so label them clearly.
Do not use color alone to show meaning	Add labels so the charts are easy to read for everyone.

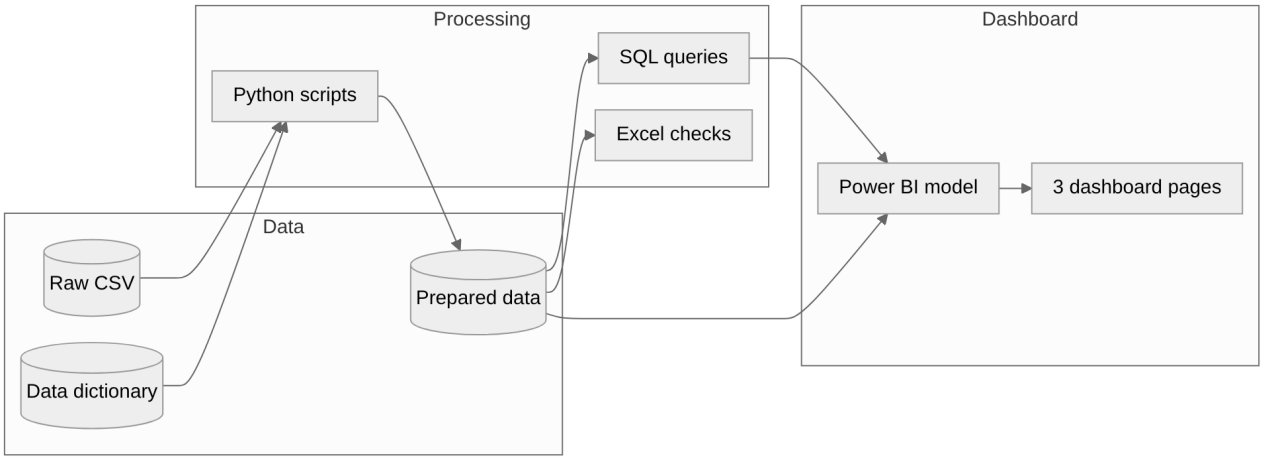
6. Tools and Integration

6.1 Tools

Tool	How we use it
Python	Load the CSV, check the data, find quality issues and clean it
SQL	Run queries to calculate the KPIs (no database is built)
Excel	Manual checks of the results
Power BI	Build the data model and the dashboard

6.2 Component diagram

What it shows: The main parts of the project and what depends on what.



The raw CSV and the dictionary feed the Python scripts, which produce the prepared data. SQL queries and Excel use the prepared data, and Power BI builds the three dashboard pages from it.

6.3 Data workflow

Step	What we do	Tool	Result
1	Load the CSV file	Python	Raw data in memory
2	Check and compare with the dictionary	Python	List of quality issues
3	Run the tests (Section 7)	Python, SQL	Pass or fail results
4	Calculate the KPIs	SQL	KPI tables
5	Build the dashboard	Power BI	3 pages
6	Compare dashboard totals with the results	Excel, Python	Checked dashboard

7. Testing

7.1 Data checks

The expected results are what the current file shows.

Check	Expected result
Number of rows	180,519
Number of columns	53
Order Item Id is unique	No repeated values
No duplicate rows	0 duplicates

Check	Expected result
Each order has one customer	0 exceptions
Sales = price x quantity	0 mismatches
Order Item Total = Sales - Discount	Matches (tiny rounding differences only)
Delivery Status matches the shipping days (not cancelled)	All rows match
Dates can be read and shipping is never before the order	0 errors

7.2 Dashboard checks

Check	Expected result
Totals match in Python, SQL, Excel and Power BI	Sales 36,784,735.01; Orders 65,752
Sales by Market add up to the total Sales	Same number
Late Delivery Rate is calculated again independently	54.83%
Each user can find their KPIs without help	Sales, Logistics / Delivery Manager and Board Viewer all confirm